SYSTEMATIC REVIEW

BJD
British Journal of Dermatology

Risks for *Staphylococcus aureus* colonization in patients with psoriasis: a systematic review and meta-analysis

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Linked Comment: Hay. *Br J Dermatol* 2017; **177**:894–895.

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Accepted for publication
22 January 2017

Funding sources
None.

Conflicts of interest
None declared.

DOI 10.1111/bjd.15366

Summary

Evidence on whether patients with psoriasis have a higher risk for staphylococcal colonization than healthy controls remains controversial. To synthesize the current literature, we performed a systematic review on the prevalence and relative risk (RR) of *Staphylococcus aureus* colonization in patients with psoriasis. We modified the QUADAS-2 instrument to assess the reporting quality of individual studies and applied random-effects models in meta-analysis. Overall we identified 21 eligible studies, of which 15 enrolled one or more comparison groups. The pooled prevalence of staphylococcal colonization in patients with psoriasis was 35·3% [95% confidence interval (CI) 25·0–45·6] on lesional skin and 39·2% (95% CI 33·7–44·8) in the nares. Patients with psoriasis were 4·5 times more likely to be colonized by *S. aureus* than healthy controls were on the skin (RR 5·54, 95% CI 3·21–9·57) and 60% more in the nares (RR 1·60, 95% CI 1·11–2·32). Cutaneous and nasal colonization by meticillin-resistant *S. aureus* also appeared higher in patients with psoriasis (pooled prevalence 8·6%) than in healthy controls (2·6%), yet the difference was not statistically significant (P = 0·74). In contrast, despite of a similar risk for nasal staphylococcal colonization (RR 0·67, 95% CI 0·38–1·18), patients with psoriasis were less likely to carry *S. aureus* on lesional skin than atopic patients (RR 0·64, 95% CI 0·40–1·02). In summarizing the current literature, we found that patients with psoriasis were at an increased risk for staphylococcal colonization compared with healthy individuals. Prospective studies on how bacterial loads correlate with disease activity can guide the clinical management of bacterial colonization while preventing the emergence of drug-resistant strains.

What's already known about this topic?

- *Staphylococcus aureus* colonizing the skin of patients with psoriasis may trigger or worsen the disease activity.

What does this study add?

- By summarizing the current literature, we found that patients with psoriasis were at a significantly increased risk of staphylococcal colonization on lesional skin and in the anterior nares compared with healthy controls.
- Patients with psoriasis also seemed to have a similar risk of staphylococcal colonization to patients with atopic diseases.

Psoriasis vulgaris (psoriasis) is a chronic inflammatory cutaneous disorder, affecting approximately 2–4% of the general population.[1] While it is well established that *Staphylococcus aureus* may exacerbate clinical symptoms of atopic dermatitis via superantigen-directed Toll-like receptor-2 pathways and selective stimulation of T-cell expression,[2,3] investigations on how

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staphylococcal superantigens link to the disease activity of psoriasis have just begun to accumulate.[4–6]

Although dysregulation of the skin microbiome and cutaneous colonization by *S. aureus* are known to perpetuate psoriasis via altered immune responses,[7,8] bacterial colonization or infection has received minimal attention in the clinical management of psoriasis.[9] This lack of guidance may have resulted from two factors: on the one hand, the focus on infections as treatment-associated adverse events rather than as a disease component in psoriasis,[10] and on the other hand, clinical observations being inconsistent about the risk of staphylococcal colonization in patients with psoriasis.[6,11–14]

To summarize the current literature and to investigate potential sources of heterogeneous observations, we performed a systematic review on published risk estimates for *S. aureus* colonization in patients with psoriasis. We also quantified risk comparisons by estimating relative risks (RRs) and absolute risk differences (RDs) when comparing patients with psoriasis with healthy controls, and with patients with atopic diseases, separately.

Materials and methods

The conduct of the literature search, critical appraisal, evidence synthesis (both qualitatively and quantitatively) and reporting closely followed the published PRISMA reporting guideline.[15]

Search strategy

We applied database-specific search terms in PubMed, Embase, Web of Science, CINAHL, SCOPUS and the Cochrane Database up to December 2014 in the first attempt (Appendix S1; see Supporting Information). Later, we used the same set of search strategies for an updated search up to February 2016. We did not constrain either search on publication types but limited the search only to indexed, peer-reviewed journals so as to ensure the quality of publications.

We screened titles and abstracts in the English language for eligibility to full-text assessment. For title-only entries, we retrieved the full texts to determine the eligibility unless the title clearly indicated its irrelevance, such as studies not in humans. When a full text was unavailable through membership subscription by author-affiliated institutions (Johns Hopkins Welch Medical Library, Baltimore MD, U.S.A.; Chang Gung University, Gueishan, Taiwan; Chang Gung Memorial Hospital, Linkou, Taiwan), we excluded it and categorized it as 'inaccessible'. For eligible abstracts without English full texts, we browsed through the tables and figures to decide whether there was sufficient information for inclusion.

Study selection and data extraction

Two reviewers (C.F.C. and T.C.W.) independently screened the titles and abstracts and we resolved inconsistencies based on group discussion with a third reviewer (S.H.L. or Y.H.H.), who also independently reviewed all titles and abstracts and further verified the final decision to retrieve full-text articles. Four reviewers (C.F.C., T.C.W., C.Y.N. and S.H.L.) then independently screened each full-text article based on three criteria: (i) whether it was an original work as a case series or a cohort of patients with psoriasis; (ii) whether the publication reported risks of staphylococcal colonization or infection in terms of prevalence estimates or incidence rates with or without a comparison group and (iii) the colonization status resulting from the conventional culture methods. We excluded review articles, case–control studies in which cases and controls were defined by colonization status, and single-case reports. During the updated search in February 2016, C.Y.N. and S.H.L. separately evaluated the updated search results, and made decisions on obtaining and assessing the eligibility of full-text articles. There were no eligible studies added to this updated search.

We collected the following information using a structured data-abstraction form: (i) article information (first author's name, year of publication and country or city where the study was conducted; (ii) study population – study setting in which patients were enrolled, number of patients included, demographic characteristics of patients, unit of comparison (patient, body site sampled or bacterial isolate); (iii) study design (cross-sectional, pre- vs. postcomparison or longitudinal survey) and (iv) risks of staphylococcal colonization or infection reported [point estimate and 95% confidence interval (CI) or SE] as ascertained by bacterial culture.

Two reviewers (C.F.C. and T.C.W.) independently extracted data from all articles that were eligible for evidence synthesis. Another two reviewers (C.Y.N. and S.H.L.) separately verified and compared the doubly entered data and reached consensus by discussion with a clinical expert on psoriasis (Y.H.H.). All investigators involved in the process of literature search and critical review have been thoroughly trained with skills for database searching and critical appraisal of the medical literature.

Reporting quality and risk-of-bias assessment

As we found that the Newcastle–Ottawa Scale was not readily applicable to the included studies,[16] we modified the QUADAS-2 instrument to assess the reporting quality of individual studies (Fig. S1; see Supporting Information).[17] Two reviewers (C.Y.N. and S.H.L.) independently evaluated each included publication, decided the degree of potential risks of bias and scored each item with 1–3 points corresponding to a high-to-low level of risk (Table S1; see Supporting Information).

Statistical analysis and meta-analysis

We extracted (or calculated) the published prevalence of staphylococcal colonization overall and, when available, by sampling body sites. We pooled original estimates, including boundary prevalence estimates (0% or 100%), using fixed-effects (FE) or random-effects (RE) models where appropriate.[18] For comparative studies, we calculated prevalence differences and prevalence ratios to approximate the RD and RR of staphylococcal colonization by comparing patients with psoriasis against healthy controls or atopic patients.

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In the meta-analysis, we employed and reported both the inverse variance (FE models) and the DerSimonian and Laird method (RE models) for comparison purposes; we also reported the degree of between-study heterogeneity based on the I^2-statistics and results of the Cochrane Q-test to justify the choice of an FE or RE model.[19] For studies with an internal comparison group (lesional vs. nonlesional) we used RE models to pool cross-site comparisons while accounting for within-person correlation. We chose prevalence ratios over odds ratios because the latter tend to overestimate the strength of associations when the outcome of interest is not a rare condition (> 10%) in the study population.[20]

We investigated potential sources of heterogeneity using several approaches. Firstly, we stratified the pooled analysis by potential confounders at the study level, including the age of the study population (adults only vs. children/adolescents or mixed age groups), levels of reporting quality and sources of study populations (inpatient, outpatient, community or 'not reported'). Secondly, we performed univariate metaregression models to identify covariates that explained a significant portion of the observed variation among studies. We also used funnel plots to examine for publication bias[19] and applied Harbord's modified test for potential small-study effects.[21] Lastly, we performed influence analysis to identify outliers using a user-written Stata program ('metainf').[22] We performed all statistical analyses in Stata 13 (StataCorp, College Station, TX, U.S.A.) at a two-tailed significance level of 0·05 unless otherwise specified.[23]

Results

We identified 581 records in the literature search, and retrieved 202 articles for further full-text assessment (Fig. 1). After excluding 181 articles for various reasons, we included 21 eligible studies for qualitative and 19 for quantitative synthesis.

Study characteristics

Nineteen of the 21 publications included were cross-sectional investigations (Table 1); the other two studies performed repeated sampling but did not achieve complete follow-up.[14,24] Therefore, we reported only baseline prevalence in Table 1 and in the following meta-analysis. The majority of studies were based in the U.S.A. (n = 9) or in European countries (n = 8). There were three main sources of patient populations, with the most common source being outpatients (n = 7).[6,12,13,25–28] Authors of these prevalence studies infrequently reported the

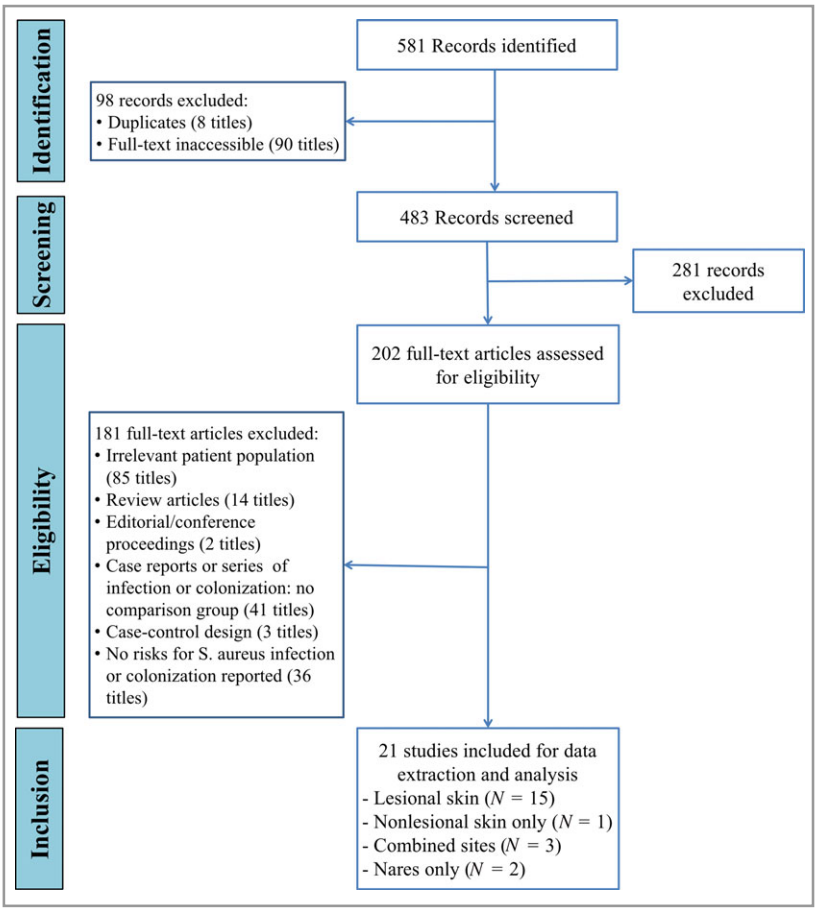

Fig 1. Flow diagram of article search and selection. Duplicates were identified manually using criteria of the same study population and results, but being published in different journals or years or with different order of authorship.

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age or sex distribution of the study population: among 12 studies of known age compositions, nine enrolled only adults aged 18 years or older.[6,7,13,27,29–32,33]

Staphylococcal colonization on the skin

Among 21 included studies, 14 reported prevalence estimates of staphylococcal colonization for skin lesions of patients with psoriasis (pooled estimate 35·3%, 95% CI 25·0–45·6), including 11 comparative studies with one or more external comparison groups and three with internal (cross-body-site) comparisons.[6,12,29]

The current meta-analysis only considered risk comparisons made across body sites and with two external control groups: healthy volunteers[6,7,13,14,26,28,30,31] and patients with atopic diseases including atopic dermatitis,[14,26,31,32,34–36] eczema[11,37] and respiratory allergy or asthma.[36] We excluded risk comparison estimates made relative to patients with other skin or immune-mediated conditions;[11,13,33] we also excluded two studies in which there were no comparison groups from the meta-analysis (Table S2; see Supporting Information).[12,27]

Eight studies reported and compared *S. aureus* colonization risks on skin of patients with psoriasis and of their healthy counterparts (Table 2). Among these studies, the prevalence of staphylococcal colonization was 36·6% on skin lesions of patients with psoriasis (RE estimate, 95% CI 18·3–54·8) and 5·1% in healthy controls (FE estimate, 95% CI 3·0–8·5). On average, patients with psoriasis consistently showed a 4·5-fold higher risk for staphylococcal colonization than healthy individuals (pooled RR 5·54, 95% CI 3·21–9·57; Fig. 2a). Despite significant between-study heterogeneities in the absolute scale, patients with psoriasis similarly had a 30% higher risk of carrying *S. aureus* than their healthy counterparts (RD 31·4%, Fig. S2a; see Supporting Information).

In contrast to the results in healthy controls, published staphylococcal colonization rates in atopic patients varied substantially across studies ($P < 0·001$ for between-study heterogeneity, Table 2), with a pooled estimate of 93·7% (95% CI 60·2–99·3), whereas the pooled estimate was 45·0% for the similarly enrolled patients with psoriasis (95% CI 31·3–58·8). When compared against atopic patients, patients with psoriasis were 36% less likely to harbour *S. aureus* on their skin lesions (RR 0·64, 95% CI 0·40–1·02; Fig. 2b), although the difference was statistically significant only in the absolute scale (RD −30·0%, 95% CI −57·0 to −2·0; Fig. S2b; see Supporting Information).

In four studies, the investigators compared colonization risks between involved vs. uninvolved skin of the same patient (Table 3).[6,12,25,29] However, to avoid introducing surveillance bias, we excluded estimates that were based on numbers of positive swabs (rather than of patients) from the statistical pooling.[25] The results showed no significant differences in the absolute colonization burden (95% CI −13·3–70·7), although lesional sites had a 1·9-fold higher likelihood of carrying *S. aureus* than uninvolved skin for a given patient with psoriasis (RR 2·89, 95% CI 1·04–8·04).[6,12]

Staphylococcal colonization in the nares

According to four studies reporting the prevalence of nasal staphylococcal carriage (Table 4),[6,13,14,31] 49·5% of patients with psoriasis (95% CI 40·0–59·0) and 26·7% of healthy participants (95% CI 10·7–42·7) carried *S. aureus* in the anterior nares. Overall, patients with psoriasis had a 60% higher risk for nasal staphylococcal colonization than healthy controls (RR 1·60, 95% CI 1·11–2·32; Fig. 3a), or by nearly 25% in the absolute scale (RD 24·6%, Fig. S3a; see Supporting Information). Meanwhile, patients with psoriasis appeared to have a similar risk to atopic patients for nasal staphylococcal colonization in both the relative (RR 0·67, 95% CI 0·38–1·18; Fig. 3b) and absolute scales (Fig. S3b; see Supporting Information).

Meticillin-resistant *Staphylococcus aureus* colonization

Because few studies had a sufficient sample size,[13,14] we pooled published risk estimates for skin with those for nasal meticillin-resistant *Staphylococcus aureus* (MRSA) colonization (Table 5). Overall, the evidence suggested a barely increased risk of MRSA colonization in patients with psoriasis vs. healthy individuals (RD 2·0%, 95% CI −9·7–13·6). Compared with atopic patients, patients with psoriasis were less likely to carry MRSA on the skin lesions (RD −6·2%) or in the nares (RD −5·4%), yet the differences were of only borderline significance (Table 5).

Quality assessment and impacts

In stratified analysis, the results of risk comparison between patients with psoriasis and healthy controls (or atopic patients) did not differ substantially from the overall findings, despite large uncertainties in certain subgroups by age (Fig. S4; see Supporting Information) and by source populations (Fig. S5). Results from low-quality reports did not deviate from those of high-quality ones but were less reliable (Fig. S6). No study-level characteristics seemed to correlate with between-study variations in meta-regression models (data not shown). While there was no statistical evidence of small-study effects as indicated by the modified Harbord's test, several studies lay outside of the 95% CI in funnel plots (Fig. S7; see Supporting Information); however, exclusion of each did not considerably alter the meta-analysis results (Fig. S8).

Discussion

We systematically identified and synthesized the current literature on staphylococcal colonization in patients with psoriasis. We showed that more than one-third of patients with psoriasis carried *S. aureus* on their skin lesions (pooled estimate 35·3%). When compared with healthy individuals, patients with psoriasis had a significantly increased risk of staphylococcal colonization either on the skin (RR 5·54) or in the nares (RR 1·60). Also, the colonization risk in psoriasis was no less

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Table 1 Summary characteristics of 21 original studies included in the current review for risks for staphylococcal colonization in patients with psoriasis, sorted by publication year

Study	Country and study period	Patient source	Study population	Age (years), mean (range)	Male (%)	*Staphylococcus aureus* colonization (%)	Quality score
Noble 1966[11]	U.K.	Inpatient	130 patients with skin diseases, including 35 with eczema, 36 with psoriasis and 20 with stasis ulcers	NA	NA	Combined 27·8%	4
Noble 1968[53]	U.K.	Inpatient and outpatient	34 psoriasis	NA	NA	Combined 26·5%	4
Marples 1973[25]	U.S.A.	Outpatient	61 psoriasis	NA	NA	Lesional 45·7%, nonlesional 26·9%[a]	5
Aly 1976[12]	U.S.A.	Outpatient	40 psoriasis	NA	NA	Lesional 20%, nasal 30%, nonlesional 13%	4
Leyden 1978[26]	U.S.A.	Outpatient	40 healthy infants; 100 infants with diaper dermatitis, 24 of whom had AD and 2 psoriasis	1 week to 15 months	NA	Lesional 50%	7
Weissman 1980[27]	U.K.	Outpatient	10 psoriasis	43 (20–75)	60%	Nonlesional 70%	5
Leung 1993[35]	U.S.A.	Unclear	56 patients with AD (42 had culture results) with four control groups: 15 healthy, nonatopic; 10 respiratory allergy; 16 psoriasis; 7 HIE syndrome	NA	NA	Lesional 56%	3
Klein 1997[34]	U.S.A.	Inpatient and outpatient	33 psoriasis; 28 AD; 33 HIV positive	NA	NA	Lesional 21%, nasal 33%	4
Sayama 1998[29]	Japan	Unclear	100 chronic plaque psoriasis	18–84	56%	Lesional 15%, nonlesional 5·6%	3
Brook 2002[37]	U.S.A.	Unclear	150 children, 23 of whom had psoriasis and 41 eczema	NA	NA	Lesional 65%	3
Flytström 2003[30]	Sweden	Inpatient and outpatient	32 untreated patients with psoriasis; 19 controls	61 (32–81)	56%	Lesional 38%	8
Tomi 2005[31]	Austria, January 2001 to March 2002	Unclear	115 patients, including 25 psoriasis, 25 AD and 25 healthy controls	54[b] (21–80)	72%	Lesional 44%, nasal 56% combined 60%	4
Nguyen 2008[13]	U.S.A., August 2007 to February 2008	Outpatient	25 psoriasis, 25 healthy controls, 50 CTCL	59·6 (28–83)	56%	Lesional 16%, nasal 44%, combined 48%	8
Balci 2009[6]	Turkey	Outpatient	50 patients and 50 sex- and age-matched healthy controls	42 ± 15·7	NA	Lesional 64%, nasal 50%, nonlesional 14%	6
Reich-Schupke 2010[32]	Germany, September 2008 to November 2008	Inpatient	715 dermatology inpatients, 50 of whom had psoriasis and 40 AD	57·1	41%	Nasal (MRSA) 0%	6
Casas 2011[7]	Germany	Unclear	39 psoriasis and 39 healthy volunteers	54 (26–75)	NA	Lesional 46%	8

(continued)

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Table 1 (continued)

Study	Country and study period	Patient source	Study population	Age (years), mean (range)	Male (%)	*Staphylococcus aureus* colonization (%)	Quality score
Graber 2011[14]	France[c]	Community	31 children with chronic dermatoses, including 7 AD and 5 psoriasis; 19 family members as controls	13·8 (10–20)	60%	Lesional 40%, nasal 40%	6
Marcus 2011[24]	U.S.A., 16 months	Inpatient	98 samples from 73 patients with psoriasis[d]	NA	NA	Lesional 32%	3
Atefi 2012[28]	Iran, 2006–2009	Outpatient	40 patients and 40 controls	43 (9–86)[e]	65%	Lesional 7·5%	5
Andersen 2013[36]	Denmark, 2008–2011	Community	2196 Danish twins, including 112 with psoriasis and 607 with atopic diseases (AD, asthma, allergy)	NA	NA	Nasal 37·5%	6
Varley 2014[33]	U.S.A.	Inpatient	548 IMID, including 285 with psoriasis and 154 with RA	48[b] (18–79)	49%	Combined 43·5%[f]	3

AD, atopic dermatitis; CTCL, cutaneous T-cell lymphoma; HIE, hyper-IgE syndrome; IMID, immune-mediated inflammatory diseases; MRSA, meticillin-resistant *Staphylococcus aureus*; NA, not available; RA, rheumatoid arthritis. [a]Results were based on 92 lesional samples taken from 61 patients and 52 nonlesional samples taken from 50 (out of 61) patients. [b]Median age. [c]Three consecutive annual summer camps in June 2005, 2006 and 2007; only findings in 2005 were shown. [d]Forty-two of whom were sampled twice or more. [e]Of patients with psoriasis; controls were age and sex matched. [f]Nares and inguinal folds were sampled.

Table 2 Results of individual and pooled prevalence estimates of staphylococcal colonization on lesional skin of patients with psoriasis vs. healthy controls (normal skin) and patients with atopic disease

	Prevalence (%)	
Study	Psoriasis	Healthy controls
Leyden 1978[26]	50·0	0
Flytström 2003[30]	37·5	5·3
Tomi 2005[31]	44·0	4·0
Nguyen 2008[13]	16	8·0
Balci 2009[6]	64·0	12·0
Casas 2011[7]	46·2	0
Graber 2011[14, a]	40·0	10·5
Atefi 2012[28]	7·5	2·5
Fixed-effects estimates	28·0	5·1
Random-effects estimates	36·6	4·3
I^2 or τ^2 (P for heterogeneity)	89·2% (P < 0·001)	0·41 (P = 0·13)
Study	**Psoriasis**	**Atopic disease**
Leyden 1978[26]	50·0	100
Leung 1993[35]	56·3	100
Klein 1997[34]	21·2	64·3
Brook 2002[37]	65·2	29·3
Tomi 2005[31]	44·0	60·0
Casas 2011[7]	46·1	100
Fixed-effects estimates	42·1	73·0
Random-effects estimates	45·0	93·7
I^2 or τ^2 (P for heterogeneity)	61·8% (P = 0·016)	7·97 (P < 0·001)

[a]Samples were taken from skin with active lesions or from hands of participants with psoriasis.

than that in atopic disease, particularly in the community cohorts.[14,36] According to a limited number of studies, patients with psoriasis had a comparable risk for MRSA colonization with healthy controls; they were less likely than atopic patients to carry these drug-resistant strains (RR 0·17, 95% CI 0·03–0·95).

Over the past decade, a growing body of evidence has shown that pathogen–host interactions involving both the innate and the adaptive immune pathways are likely to be among the driving forces underlying the dysfunctional T helper (Th)1/Th17 immune axis in psoriasis.[38–41] An early finding of V-B element-bearing T cells enriched in psoriatic lesions led to the speculative role of microbial superantigens in the pathogenesis of psoriasis.[42,43] Such speculation was later supported by observation of an increased prevalence of staphylococcal superantigen in the lesional skin of patients with psoriasis compared with skin of healthy controls.[6]

Using culture-independent techniques, researchers have also shown that the relative abundance of *S. aureus*, along with other Firmicutes[44] or other types of *Staphylococcus*,[45] was increased in lesional skin compared with nonlesional sites of patients with psoriasis (or with healthy skin), although there were reports to the contrary.[46] As the consensus on how to

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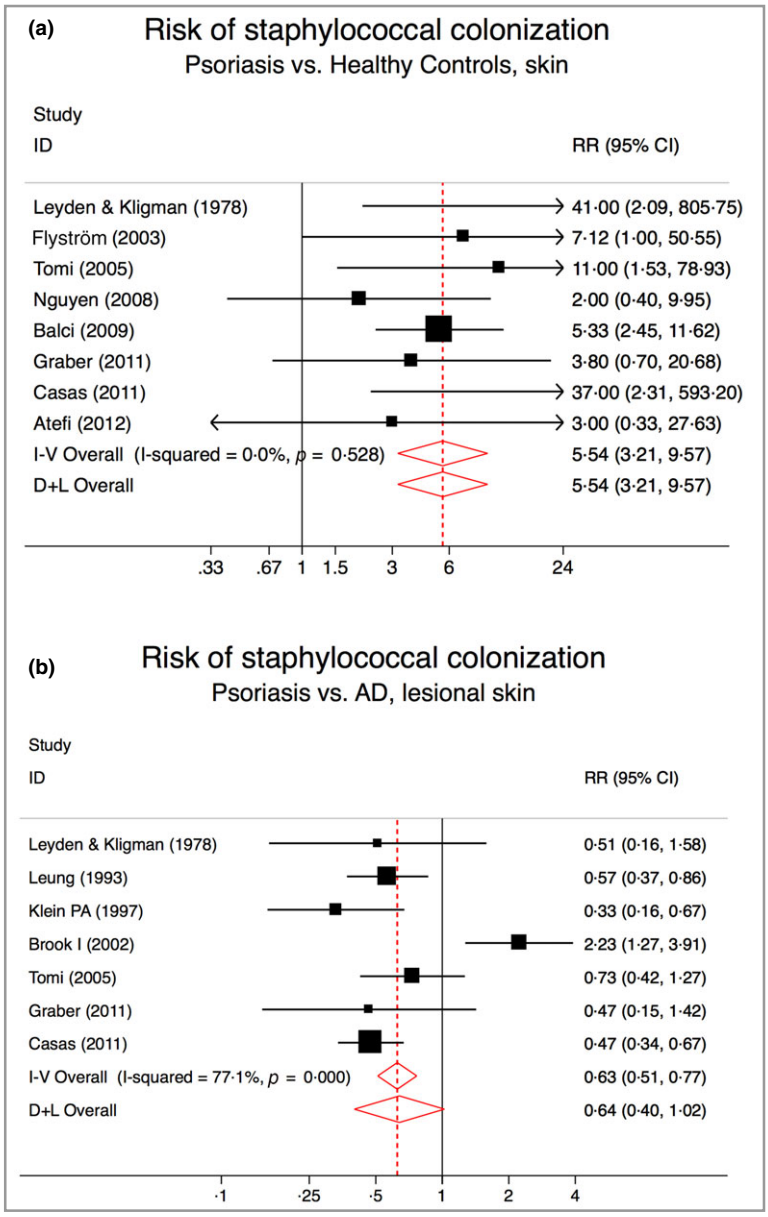

Fig 2. Risk of staphylococcal colonization in the skin. (a) Psoriasis vs. healthy controls. (b) Psoriasis vs. atopic dermatitis (AD). RR, relative risk; CI, confidence interval.

Table 3 Results of meta-analysis on risks for staphylococcal colonization in lesional vs. nonlesional skin in patients with psoriasis

			Colonized/total (%)			
Study	Age (years)	Male (%)	Lesional	Nonlesional	RD (95% CI)	RR (95% CI)
Aly 1976[12]	NA	NA	8/40 (20)	5/40 (13)	7·5% (−8·6–23·6)	1·60 (0·57–4·47)
Sayama 1998[29]	Range 18–84	56%	15/100 (15)	5/89 (6)	9·4% (0·9–17·9)	2·67 (1·01–7·05)
Balci 2009[6]	Mean 42·2 ± 15·7	NA	32/50 (64)	7/50 (14)	50% (33·6–66·4)	4·57 (2·23–9·37)
Random-effects pooled estimates (95% CI)			32·6% (4·5–60·8)	9·2% (3·6–14·9)	21·8% (−2·7–46·3)	2·95 (1·61–5·42)

CI, confidence interval; NA, not available; RD, risk difference; RR, relative risk.

conduct microbiome research on dermatological disorders has just begun to build,[47] we expect to see a more and more comprehensive characterization of the skin microbial community that is associated with particular diseases at the onset, in remission or in relapse.

In comparison with a recently published systematic review on staphylococcal colonization in patients with atopic dermatitis,[48] original epidemiological investigations of the psoriasis population have been relatively few and poorly conducted or reported (Table S1; see Supporting Information). Despite the

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Table 4 Results of original and pooled prevalence of nasal staphylococcal colonization in patients with psoriasis vs. healthy controls and patients with atopic disease

	Prevalence (%)	
Study	Psoriasis	Healthy
Tomi 2005[31]	56·0	8·0
Nguyen 2008[13]	44·0	28·0
Balci 2009[6]	50·0	34·0
Graber 2011[14]	40·0	42·1
Fixed-effects estimates	49·5	22·3
Random-effects estimates	49·5	26·7
I^2 (P for heterogeneity)	0·0% (P = 0·82)	77·7% (P = 0·004)
Study	Psoriasis	Atopic disease
Klein 1997[34]	33·3	78·6
Tomi 2005[31]	56·0	88·0
Graber 2011[14]	40·0	85·7
Andersen 2013[36]	35·7	29·0
Fixed-effects estimates	38·1	36·5
Random-effects estimates	39·0	69·7
I^2 (P for heterogeneity)	22·6% (P = 0·28)	97·6% (P < 0·001)

fact that more than half of the studies recruited at least one external comparison group, most investigators did not provide sufficient details on the participant selection and enrolment procedures, and whether culture results were interpreted independently of clinical diagnoses (Fig. S1; see Supporting Information). Both, if unduly executed, would result in biased assessments of absolute and relative risks.[17]

As *S. aureus* can reside among the human host's skin and nasal microbiota in the general population, with a prevalence of asymptomatic nasal colonization approximating 20–30%,[49,50] an adequately selected (or matched) control group has a pivotal role in deriving valid risk comparisons. Well-known confounders, such as male sex, younger age, race/ethnicity and chronic medical conditions including diabetes mellitus and obesity,[51] should therefore be adjusted for to provide valid and reliable risk estimates. Yet none of the studies in the current review addressed these potential confounding effects.

Among the original studies included in the current review, the study populations were poorly characterized and the patient spectrum varied from study to study, possibly

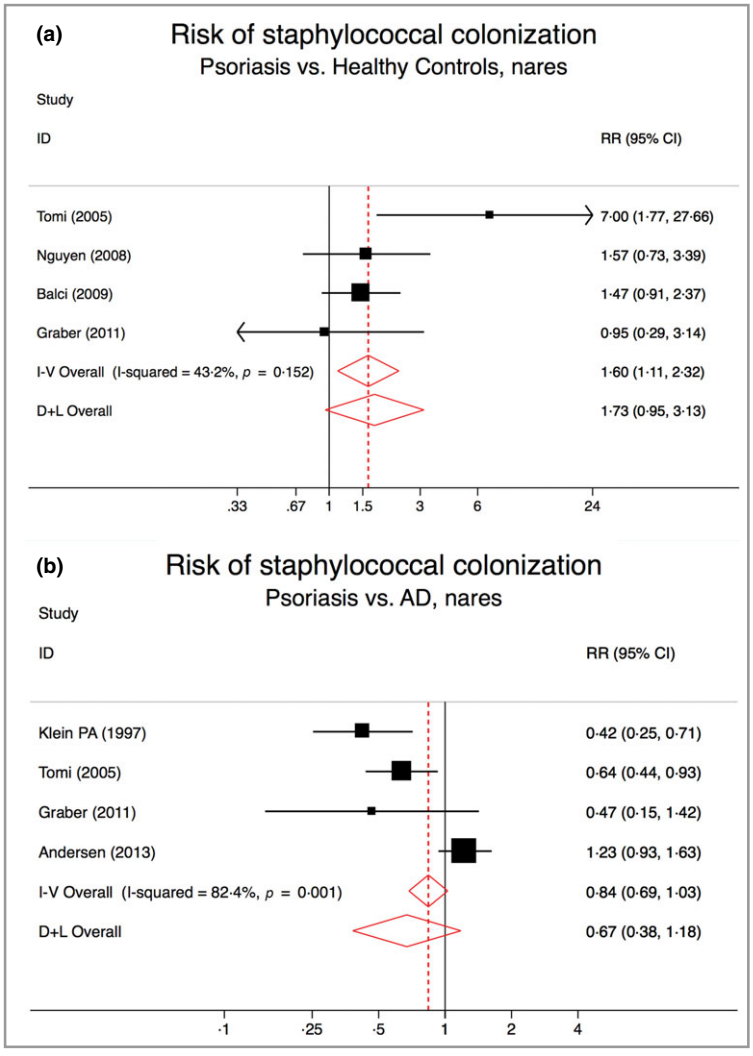

Fig 3. Risk of staphylococcal colonization in the nares. (a) Psoriasis vs. healthy controls. (b) Psoriasis vs. atopic dermatitis. RR, relative risk; CI, confidence interval.

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Table 5 Individual and pooled prevalence, and risk comparisons of meticillin-resistant *Staphylococcus aureus* colonization in patients with psoriasis, patients with atopic disease and healthy controls

Study	Sampling sites	Colonized/ total	Prevalence (%) (95% CI)	Colonized/ total	Prevalence (95% CI)	RD (95% CI)	RR (95% CI)
		Psoriasis		Healthy controls			
Nguyen 2008[13]	Nares and lesion	3/25	12·0 (2·5–31·2)	2/25	8·0 (1·0–26·0)	4 (−12·6–20·6)	1·50 (0·27–8·22)
Graber 2011[14]	Lesion	0/5	0·0 (0·0–52·2)	0/19	0·0 (0·0–17·6)	0 (−23·1–23·1)	–
Graber 2011[14]	Nares	0/5	0·0 (0·0–52·2)	0/19	0·0 (0·0–17·6)	0 (−23·1–23·1)	–
RE estimates			8·6 (2·8–23·4)		2·6 (0·3–21·1)	2 (−9·7–13·6)	–
		Psoriasis		Atopic disease			
Klein 1997[34]	Lesion	0/33	0·0 (0·0–10·6)	2/28	7·1 (0·9–23·5)	−7·1 (−18·1–3·8)	0·17 (0·01–3·41)
Graber 2011[14]	Lesion	0/5	0·0 (0·0–52·2)	0/7	0·0 (0·0–41·0)	0 (−27·8–27·8)	–
FE estimates			0		5·7 (1·4–20·2)	−6·2 (−16·4–4·0)	–
Klein 1997[34]	Nares	0/33	0·0 (0·0–10·6)	2/28	7·1 (0·9–23·5)	−7·1 (−18·1–3·8)	0·17 (0·01–3·41)
Reich-Schupke 2010[32]	Nares	0/50	0·0 (0·0–7·1)	2/40	5·0 (0·6–16·9)	−5 (−12·8–2·8)	0·16 (0·01–3·26)
Graber 2011[14]	Nares	0/5	0·0 (0·0–52·2)	0/7	0·0 (0·0–41·0)	0 (−27·8–27·8)	–
FE estimates			0		5·3 (2·0–13·4)	−5·4 (−11·6–0·8)	0·17 (0·02–1·38)
Overall RE estimates[a]			0		5·5 (2·5–11·6)	−5·6 (−10·9–−0·3)	0·17 (0·03–0·95)

CI, confidence interval; FE, fixed effects; RD, risk difference; RE, random effects; RR, relative risk. [a]No site-specific differentials in absolute or relative risk differences between patients with psoriasis and atopic dermatitis (*P* for between-group heterogeneity = 0·90).

contributing to the large between-study variation observed in the prevalence estimates and risk comparisons. For example, specific subtypes of psoriasis (or of atopic diseases), disease duration and current or previous therapeutic agents were important clinical variables that might be associated with individual risks for bacterial colonization. Despite the fact that most investigators reported that they collected skin swabs before treatment at the outpatient clinic or on the first day of admission into the hospital, inadequate reporting has precluded possible subgroup analysis to identify further any potential prognostic or protective factors for staphylococcal colonization in patients with psoriasis.

Moreover, the number of sampling swabs collected per patient often outnumbered that from a healthy control. Such oversampling bias could be avoided by defining a fixed number of sampling body sites a priori for both patients and healthy controls. When different numbers of swabs per person were collected (as was typical in retrospective clinical studies),[24,25] reporting of culture findings at the person level, rather than at the swab level, might reduce the risk for surveillance bias (although not completely).

We also caution that the study findings should be interpreted in light of a number of methodological limitations associated with the review process. All risk comparisons considered were cross-sectional by design; when the authors reported both pre- and post-treatment prevalence rates, we included only the former in the current review and meta-analysis. How bacterial colonization in patients with psoriasis evolved over time, with or without clinical treatment, was beyond the scope of the current review. In addition, the colonization prevalence of *S. aureus* varied over the sampling body sites, but few studies reported site-specific results.[14,37] Although we recognized that *S. aureus* might show preference for specific skin types (dry, wet or oily),[52] our results pertained only to an overall assessment of staphylococcal colonization in patients with psoriasis.

The majority of the study populations considered were hospital-based outpatients. Future population-based, community surveys are required to provide further reliable evidence on whether patients with psoriasis serve as a hidden reservoir for the emergence of community-associated MRSA.[31,50] Lastly, much as this review was limited to the published literature and there were potential publication biases, we did not find evidence that was suggestive of this bias based on funnel plots, Harbord's modified test or influence analysis.

In conclusion, based on the current literature, we have demonstrated that patients with psoriasis have an increased risk of *S. aureus* colonization on the skin and in the anterior nares compared with healthy individuals. Such heightened risk for cutaneous bacterial burden is no less than that observed in atopic patients. While more mechanistic research is needed to investigate the aetiological role of *S. aureus* in the natural history of psoriasis, prospective investigations that seek to inform clinicians of optimal practices to constrain bacterial colonization and to prevent the emergence of drug-resistant strains are also warranted.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1. Database-specific search strategies used in the current study.

Fig S1. Quality of reporting of individual studies.

Fig S2. Forest plots showing individual and pooled risk differences in staphylococcal colonization on skin.

Fig S3. Forest plots showing individual and pooled risk differences in staphylococcal colonization in the anterior nares.

Fig S4. Stratified analysis by age of study population.

Fig S5. Stratified analysis by source of study population.

Fig S6. Stratified analysis by levels of reporting quality.

Fig S7. Funnel plots to examine for potential publication bias.

Fig S8. Results of influence analysis.

Table S1. Risk-of-bias assessment in individual studies included in the current review by three reporting domains.

Table S2. Summary of studies reporting staphylococcal colonization rates in patients with psoriasis by sampling site and comparison group.

Video S1. Author video.

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